

3. Univariable Probability distribution and its Mathematical Expectation

Page No.

Date. 30/12/24

• Discrete Random variable:- A random variable x is said to be discrete if it takes finite & countable infinite values.

• Probability Mass function (pmf): Let x be a discrete random variable defined on sample space Ω .

Consider $x = \{x_1, x_2, \dots, x_n\}$ be range set of x we assign $P_i = P[x = x_i]$ is called as probability mass function (p.m.f) if it satisfy the following condition.

(i) $P_i \geq 0$; $\forall i$

(ii) $\sum_{i=1}^n P_i = 1$

Ex:- Q1] A symmetric die is rolled and number on uppermost face is noted. find its probability distribution.

$\Rightarrow \Omega = \{1, 2, 3, 4, 5, 6\}$

x : No. of uppermost face is noted

$x = \{1, 2, 3, 4, 5, 6\}$

$P(x=1) =$

The probability distribution is,

x	1	2	3	4	5	6	
$P(x=x)$	$P(x=1)$	$P(x=2)$	$P(x=3)$	$P(x=4)$	$P(x=5)$	$P(x=6)$	
$P(x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	

Q8. For the following probability distribution of x :-

x	0	1	2	3	4	
$P(x=x)$	k	3k	5k	2k	k	

1) Find value of k.

2) find $P(x \geq 2)$, $P(x < 3)$, $P(x \leq 1)$

⇒ By axioms of p.m.f

$$\sum_{i=1}^n P(x=x_i) = 1$$

$$P(x=0) + P(x=1) + P(x=2) + \dots + P(x=4) = 1$$

$$k + 2k + 3k + 5k + k = 1$$

$$12k = 1$$

$$k = 1/12$$

The probability distribution is given by,

x	0	1	2	3	4	
$P(x=x)$	$1/12$	$3/12$	$5/12$	$2/12$	$1/12$	

$$\begin{aligned}\textcircled{ii} \quad P(X \geq 2) &= P(X=2) + P(X=3) + P(X=4) \\ &= \frac{5}{12} + \frac{2}{12} + \frac{1}{12} \\ &= \frac{8}{12}\end{aligned}$$

$$\begin{aligned}\textcircled{iii} \quad P(X < 3) &= P(X=0) + P(X=1) + P(X=2) \\ &= \frac{1}{12} + \frac{3}{12} + \frac{5}{12} \\ &= \frac{9}{12}\end{aligned}$$

$$\begin{aligned}\textcircled{iv} \quad P(X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{1}{12} + \frac{3}{12} \\ &= \frac{4}{12}\end{aligned}$$

Ex3 Whether the following function can be regarded as the p.m.f for given values of x :-

①

x	1	2	3	4
$P(x)$	0.2	0.4	0.3	0.5

Here, $P_1 = 0.2 > 0$

$P_2 = 0.4 > 0$

$P_3 = 0.3 > 0$

$$P_3 = 0.3 > 0$$

$$P_4 = 0.5 > 0$$

Here all P_i values are ~~graph~~ greater than zero.

$\therefore$ 1st condition is satisfied

$$\sum_{i=1}^n P_i = 1$$

$$\text{L.H.S} = P_1 + P_2 + P_3 + P_4$$

$$= 0.2 + 0.4 + 0.3 + 0.5$$

$$= 1.4$$

$$\text{L.H.S} \neq \text{R.H.S}$$

Here IInd condition is not satisfied

$\therefore$ given probability distribution is not a p.m.f.

①

x	-1	2	3	4
P(x)	0.5	-0.3	0.3	0.5

Here,

$$P(X=-1) = 0.5 \geq 0$$

$$P(X=2) = -0.3 \leq 0$$

$\therefore$ 1st condition is not satisfied therefore the given probability distribution is not a p.m.f.

Page No. _____
Date. ____/____/____

$$(iii) \quad P(x) = \frac{x^2}{14} \quad x=1,2,3$$

$$= 0, \text{ otherwise}$$

x	1	2	3	
P(x)	1/14	4/14	9/14	

Here, $P(x=1) = \frac{1}{14} \geq 0$

$P(x=2) = \frac{4}{14} \geq 0$

$P(x=3) = \frac{9}{14} \geq 0$

Here, 1st condition is satisfied.

$$\sum_{i=1}^n P_i = 1$$

$$\begin{aligned} \text{L.H.S} &= P(x=1) + P(x=2) + P(x=3) \\ &= \frac{1}{14} + \frac{4}{14} + \frac{9}{14} \end{aligned}$$

$$= \frac{14}{14}$$

$$= 1$$

$\therefore \text{L.H.S} = \text{R.H.S}$

2nd condition is satisfied as well. Therefore the given probability distribution is p.m.f

$$4) P(x) = \frac{x-1}{2} \quad ; x = 0, 1, 2, \dots$$

$$= 0 \quad ; \text{otherwise}$$

x	0	1	2
$P(x)$	$-\frac{1}{2}$	0	$\frac{1}{2}$

Here $P(x=0) = -\frac{1}{2} < 0$ thus 1st condition is not satisfied.

Therefore, given probability is not a p.m.f.

0 Cumulative distribution function (c.d.f) :- let x be a discrete random variable with values $x_1, x_2, \dots, x_n$ & related probabilities $p_1, p_2, \dots, p_n$ then its cumulative distribution function is given by,

$$F(x) = P(x \leq x)$$

Properties of distribution function (c.d.f) :-

1) $0 \leq F(x) \leq 1$

② $F(x)$ is non decreasing function
 $F(x_1) \leq F(x_2)$ or $F(x_2) \leq F(x_3)$

③ $F(-\infty) = 0$
 $F(\infty) = 1$

④ $P(a \leq x \leq b) = P(x \leq b) - P(x \leq a)$
 $= F(b) - F(a)$

⑤ $P(x \geq b) = 1 - P(x < b)$

9] Consider the following probability distribution of discrete random variable (x)

① Obtain the c.d.f of x

x_i	1	2	3	4	5
P_i	0.1	0.2	0.3	0.2	0.2

$\Rightarrow$ The c.d.f of x given by,

x_i	1	2	3	4	5
P_i	0.1	0.2	0.3	0.2	0.2
$F(x)$	0.1	0.3	0.6	0.8	1

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.1 & 1 \leq x \leq 2 \\ 0.3 & 2 \leq x \leq 3 \\ 0.6 & 3 \leq x < 4 \\ 0.8 & 4 \leq x < 5 \\ 1 & x \geq 5 \end{cases}$$

$$p.d.f = P_i = ?$$

Ex 2] The following is the cumulative distribution function of a discrete random variable

x	-3	-1	0	1	2	3	5	8	
$F(x)$	0.1	0.3	0.45	0.65	0.75	0.9	0.95	1	
$P(X=x)$	0.1	0.2	0.15	0.20	0.1	0.15	0.05	0.05	

1. find the p.m.f of x

i. $P(0 < x < 2)$

iii. $P(1 \leq x \leq 3)$

iv. $P(-3 < x \leq 2)$

v. $P(-1 \leq x < 1)$

vi. $P(X = \text{even})$

vii. $P(x > 2)$

viii. $P(x \geq 3)$

$$\textcircled{a} \quad P(0 < x < 2) := P(x=1) \\ = \boxed{0.2}$$

$$\textcircled{a} \quad P(1 \leq x \leq 3) = P(x \leq 3) - P(x \leq 1) \quad \text{---} \quad (P(0 < x < 5) = 1) \\ = F(3) - F(1) \\ = 0.9 - 0.65 \\ = \boxed{0.25}$$

$$\textcircled{a} \quad P(-3 < x \leq 2) := P(-1 \leq x \leq 2) \\ = P(x \leq 2) - P(x \leq -1) \\ = F(2) - F(-1) \\ = 0.75 - 0.3 \\ = \boxed{0.45}$$

$$\textcircled{a} \quad P(-1 \leq x < 1) = P(-1) + P(0) \\ = 0.2 + 0.15 \\ = 0.35$$

$$\textcircled{vi} \quad P(x = \text{even}) = P(x=0) + P(x=2) + P(x=4) \\ = 0.15 + 0.1 + 0.05 \\ = \boxed{0.3}$$

$$\textcircled{vii} \quad P(x > 2) = P(x=3) + P(x=5) + P(x=8) \\ = 0.15 + 0.05 + 0.05 \\ = \boxed{0.25}$$

$$\textcircled{viii} P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - P(X \leq 2)$$

$$= 1 - F(2)$$

$$= 1 - 0.75$$

$$= \boxed{0.25}$$

Mathematical Expectation:- Let x be a discrete random variable taking value $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$ respectively, the mathematical expectation of x is denoted by $E(x)$

$$E(x) = x_1 \cdot p_1 + x_2 \cdot p_2 + \dots + x_n \cdot p_n$$

$$= \sum_{i=1}^n x_i p_i$$

Ex1] Obtain expectation of random variable x with the following probability distribution

x_i	1	3	5	6
$P(x_i)$	0.1	0.2	0.4	0.3

Solutn:-

x_i	$P(x_i)$	$x_i P(x_i)$
1	0.1	0.1
3	0.2	0.6
5	0.4	2
6	0.3	1.8

$$\sum x_i P(x_i) = \underline{\underline{4.5}}$$

• Expectation of a function of a Random variable:- If $y = g(x)$ is a function of random variable x then y is also a random variable with the same probability distribution of x .

$$E(y) = E(g(x)) = \sum g(x) P(x)$$

Q2]

x	0	1	2	3
$P(x)$	0.3	0.3	0.4	

$$y = 2(x) + 3$$

$$= 2(0) + 3$$

$$= 3$$

$$y = 2(1) + 3$$

$$y = 5$$

$$y = 2(2) + 3$$

$$y = 7$$

$$E(y) = \sum y P(x_i)$$

$$= 3 \times (0.3) + 5 \times (0.3) + 7 \times (0.4)$$

$$= 0.9 + 1.5 + 2.8$$

$$E(y) = 5.2$$

- Theorem 1: Expected value of the constant is the constant itself i.e. $E(C) = C$

⇒ Proof :-

Let $\{x_i, p_i\}$; $i = 1, 2, \dots, n$ denote the probability distribution of a discrete r.v. X .
Let $g(x) = C$

$$\begin{aligned} E(C) &= E(g(x)) \\ &= \sum g(x_i) \cdot p_i \\ &= C \sum p(x_i) \quad \text{--- } (C \cdot \sum p(x_i) = 1) \\ &= C(1) \\ &= C \end{aligned}$$

- Theorem 2: Effect of change of origin and scale on $E(X)$

(i) $E(X + b) = E(X) + b$

(ii) $E(ax) = aE(x)$

(iii) $E(ax + b) = aE(x) + b$

⇒ Proof :- $\Rightarrow g(x) = x + b$

$$\begin{aligned} E[g(x)] &= \sum g(x_i) p_i \\ &= \sum (x_i + b) p_i \\ &= \sum x_i p_i + \sum b p_i \\ &= E(x) + b \end{aligned}$$

(ii) $g(x) = (ax)$

$$E[g(x)] = \sum g(x_i) p_i$$

$$= \sum (ax_i) p_i$$

$$= a \sum (x_i p_i)$$

$$E[g(x)] = a E(x)$$

(3) $g(x) = a E(x) + b$

$$E[g(x)] = \sum g(x_i) p_i$$

$$= \sum (ax_i + b) p_i$$

$$= a \sum (x_i p_i) + \sum (p_i b)$$

$$= a E(x) + b \sum p_i$$

$$= a E(x) + b \cdot 1$$

$$E[g(x)] = a E(x) + b$$

◦ Variance of random variable :- Let x be a distribution $\{ \sum x_i, p_i \} = i = 1, \dots, n$. $V(x)$ denoted by σ^2 is defined as

$$\sigma^2 = V(x) = E[x - E(x)]^2$$

◦ Note :-

(i) $E(x) = \mu$

(ii) The above formula for computational convenience the following simplification is a use

$$\sigma^2 = E(x - \mu)^2$$

$$\therefore \sigma^2 = \sum_{i=1}^n (x_i - \mu)^2 p_i$$

$$\sigma^2 = \sum_{i=1}^n (x_i - 2x_i\mu + \mu^2) P_i$$

Page No.

Date. / /

using, $(a-b)^2 = a^2 - 2ab + b^2$

$$\therefore \sigma^2 = \sum_{i=1}^n x_i^2 P_i - (2x_i\mu) P_i + \mu^2 P_i$$

$$= \sum_{i=1}^n x_i^2 P_i - 2\mu \sum_{i=1}^n x_i P_i + \mu^2 \sum_{i=1}^n P_i$$

$$= E(x^2) - 2\mu E(x) + \mu^2$$

$$= E(x^2) - 2\mu \times \mu + \mu^2$$

$$= E(x^2) - \mu^2$$

$$= [E(x^2) - E(x)]^2$$

Remark:-

(i) $\text{var}(x) \geq 0$

$$E(x^2) - [E(x)]^2 \geq 0$$

$$E(x^2) \geq [E(x)]^2$$

(ii) variance of constant is zero

$$\text{var}(c) = 0$$

Q] Compute variance of x for the following probability distribution:-

(i) $P(x) = \frac{x}{30}$ $\therefore x = 0, 1, 2, 3, 4$

$$\rightarrow V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \sum x_i^2 P(x_i)$$

$$= \sum x_i^2 \left(\frac{x_i}{30} \right)$$

$$= \frac{1}{30} \sum x_i^3$$

$$= \frac{1}{30} (0^3 + 1^3 + 2^3 + 3^3 + 4^3)$$

$$= \frac{1}{30} (0 + 1 + 8 + 27 + 64)$$

$$= \frac{1}{30} (354)$$

$$\begin{array}{r} 17 \\ 81 \\ \hline 256 \\ 354 \\ \hline 354 \end{array}$$

Remark :-

(i) $\text{var}(x) \geq 0$

$$E(x^2) - [E(x)]^2 \geq 0$$

$$E(x^2) \geq [E(x)]^2$$

(ii) variance of constant is zero

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Q] Compute variance of x for the following probability distribution :-

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$$= \frac{1}{30} (354)$$

$$\begin{array}{r}
 1 \\
 17 \\
 81 \\
 \hline
 256 \\
 1 \\
 \hline
 354 \\
 \hline
 354
 \end{array}$$

$$E(x) = \sum x_i p_i$$

$$= \sum x_i \left[\frac{x_i^2}{30} \right]$$

$$= \frac{1}{30} \sum x_i^3$$

$$= \frac{1}{30} \sum (0^3 + 1^3 + 2^3 + 3^3 + 4^3)$$

$$= \frac{1}{30} (0 + 1 + 8 + 27 + 64)$$

$$= \frac{100}{30}$$

$$= 3.3$$

$$[E(x)]^2 = (3.3)^2$$

$$= 10.89$$

$$V(x) = E(x^2) - [E(x)]^2$$

$$= 11.8 - 10.89$$

$$V(x) = 0.91$$

[wrong answer]
[skip that part]

$$= 11.8 - \left(\frac{10}{3} \right)^2$$

$$V(x) = 0.68$$

[correct answer]

• Effect of the change of origin and change of scale of variance :-

$$i) \text{var}(x+b) = \text{var}(x) = \sigma^2$$

$$\begin{aligned} \text{var}(x) &= E[x - E(x)]^2 \\ &= E[x+b - E(x+b)]^2 \\ &= E[x+b - E(x) - b]^2 \\ &= E[x - E(x)]^2 \\ &= \text{var}(x) \\ &= \sigma^2 \end{aligned}$$

$$\begin{aligned} ii) \text{var}(ax) &= a^2 \text{var}(x) = a^2 \sigma^2 \\ &= a^2 E[x - E(x)]^2 \\ &= a^2 E[x+b - E(x+b)]^2 \\ &= a^2 E[x+b - E(x) - b]^2 \\ &= a^2 E[x - E(x)]^2 \\ &= a^2 \text{var}(x) \end{aligned}$$

[wrong]

$$\begin{aligned} &= E[ax - E(ax)]^2 \\ &= E[ax - aE(x)]^2 \\ &= E[a(x - E(x))]^2 \\ &= a^2 E[x - E(x)]^2 \end{aligned}$$

[correct]
[Refer this]

$$\text{var}(ax) = a^2 \sigma^2$$

$$iii) \text{var}(ax+b) = a^2 \sigma^2$$

$$\sigma^2 = \text{variance} = \sigma^2 / \text{var}(x)$$

$$(i) \text{var}(x) = E(x - E(x))^2$$

$$(ii) \text{var}(x) = E(x - \mu)^2$$

$$(iii) \text{var}(x) = E(x^2) - [E(x)]^2$$

where;

$$E(x^2) = \sum x_i^2 p(x_i)$$

$$E(x) = \sum x_i p_i$$

$$\begin{aligned} (iii) \text{var}(ax+b) &= E(x - E(x))^2 \\ &= E(ax+b - E(ax+b))^2 \\ &= E[ax+b - aE(x) - b]^2 \\ &= E[a(x - E(x))]^2 \\ &= a^2 E(x - E(x))^2 \\ &= a^2 \sigma^2 \end{aligned}$$

Q1) Mean = 5

var = 3

find $\rightarrow$ 1) $2x - 5$

2) $3 - 7x$

3) $\frac{x+1}{2}$

(1) $y = 2x - 5$

$$E(y) = 2E(x) - 5$$

$$= 2(5) - 5$$

$$\begin{aligned}
 &= 5 \\
 \text{var}(x) &= V(x) - 5 \\
 &= 3 - 0 \\
 &= \boxed{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad 3 - 7x &= E(y) \\
 E(y) &= 3 - 7E(x) \\
 &= 3 - 7(5) \\
 &= 3 - 35 \\
 &= \boxed{-32}
 \end{aligned}$$

$$\begin{aligned}
 \text{var}(y) &= 0 - 7(3) \\
 &= \boxed{-21}
 \end{aligned}$$

$$\text{(iii)} \quad \frac{x+1}{2} = E(y)$$

$$\begin{aligned}
 E(y) &= \frac{E(x) + 1}{2} \\
 &= \frac{5 + 1}{2} \\
 &= 3//
 \end{aligned}$$

$$\text{var}(y) = \frac{\text{var}(x) + 0}{0}$$

2 0/

• Moment of Random variable:-

1. Raw moments (Moments about origin i.e zero) :- The r^{th} Raw moment of x is defined as the r^{th} moment about 0

• It is denoted by μ_r'
 • Hence $\mu_r' = \mu_r' (0) = E(x)^r = \sum_{i=1}^n x_i^r p_i \quad i=1, 2, \dots$

$$\mu_1' = E(x) = \text{mean}$$

$$\mu_2' = E(x^2) = \sum x_i^2 p_i$$

$$\mu_3' = E(x^3) = \sum x_i^3 p_i$$

$$\mu_4' = E(x^4) = \sum x_i^4 p_i$$

2. Central Moment (Moment about the Arithmetic mean) :-

→ The r^{th} central moment of x is defined as the r^{th} moment of x about $E(x)$. It is denoted by μ_r

$$\mu_r = \mu_r' = E[x - E(x)]^r$$

$$= \sum_{i=1}^n [x_i - E(x)]^r p_i \quad r=1, 2, 3, 4, \dots$$

1st central moment is always zero

$$\mu_2 = E(x - E(x))^2 = \text{var}(x) = \sigma^2$$

o Relation between Raw moment and central moment:-

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - \mu_1'^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

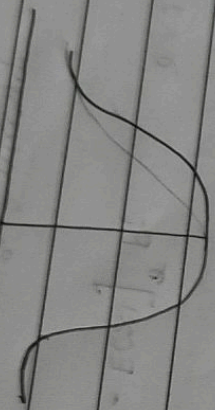
o Measure of skewness and kurtosis based on moment:-

1. Coefficient of skewness (γ_1): Skewness means the lack of symmetry of the probability

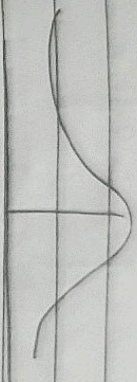
$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{\mu_3}{\mu_2^{1.5}}$$

$$\text{where } \mu_2 = 6^2$$

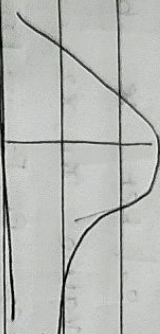
$\gamma_1 = 0$, the distribution is symmetric



$\gamma_1 > 0$, the distribution is positively skewed



$\gamma_1 < 0$; the distribution is negatively skewed



1 Coefficient of kurtosis (γ_2) :

Kurtosis means peakedness of the distribution

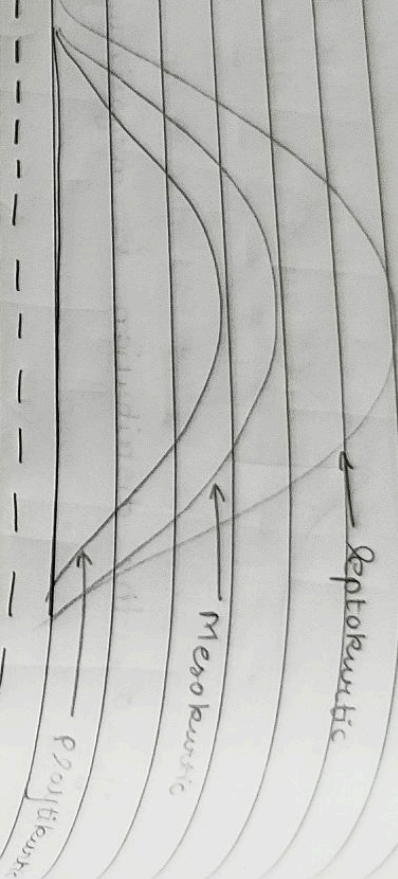
$$\gamma_2 = \frac{\mu_4 - 3\mu_2^2}{\mu_2^2}$$

γ_2 is also called the "Excess kurtosis"

$\gamma_2 = 0$ the distⁿ is mesokurtosis

$\gamma_2 > 0$ the distⁿ is leptokurtosis

$\gamma_2 < 0$ the distⁿ is platykurtosis



r^{th} Factorial Moment -] let x be a discrete r.v. taking value $x_1, x_2, \dots, x_n$ with respective probabilities $P_1, P_2, \dots, P_n$

The r^{th} factorial moment of x is denoted by $\mu_{(r)}'$

$$\mu_{(r)}' = E(x^{(r)}) \\ = E(x(x-1) \dots (x-r+1))$$

$$= \sum_{i=1}^n x_i(x_i-1) \dots (x_i-r+1) P_i \quad \text{--- (i=1,2,3)}$$

$$\mu_{(1)}' = \mu_1'$$

$$\mu_{(2)}' = \mu_2' - \mu_1'^2$$

$$\mu_{(3)}' = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_{(4)}' = \mu_4' - 6\mu_3'\mu_1' + 11\mu_2'^2 - 6\mu_1'^4$$

• Moment Generating Functions (MGF) :-

Definition: Suppose x is r.v with p.f. $P(x)$ then the moment generating function of x is denoted by $M_x(t)$ and it is defined as,

$$M_x(t) = E(e^{tx}) = \sum e^{tx} P(x)$$

$M_x(t)$ can be express in power of t

$$M_x(t) = E(e^{tx})$$

$$= E\left(1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots\right)$$

$$= 1 + tE(x) + \frac{t^2}{2!} E(x^2) + \frac{t^3}{3!} E(x^3) + \dots$$

$$= 1 + \mu_1' + \frac{t^2}{2!} \mu_2' + \frac{t^3}{3!} \mu_3' + \dots$$

0 Properties :-

1) $M_X(0) = 1$

$M_X(t) = E(e^{tx})$

$t = 0$

$M_X(0) = E(e^{0 \cdot x})$

$= E(e^0)$

$M_X(0) = E(1) = 1 \quad (\because e^0 = 1)$

$M_X(0) = 1$

2) Effect of change of origin & scale -

Result :- If a r.v. x has M.G.F. $M_X(t)$ then M.G.F. of $x+a$ is $M_{X+a}(t) = e^{at} M_X(t)$

$M_X(t) = E(e^{tx})$

$M_{X+a}(t) = E(e^{t(x+a)})$

$M_{X+a}(t) = E(e^{tx + ta})$

$= E(e^{tx} \cdot e^{ta})$

$= e^{ta} E(e^{tx})$

$M_{X+a}(t) = e^{ta} M_X(t)$

3) $M_X(t) = M_X(t)$

$M_X(t) = E(e^{tx})$

$M_{cx}(t) = E(e^{t(cx)})$

$= E(e^{ctx})$

$= E(e^{ct} \cdot e^{tx})$

$= 1$

$x^a \cdot x^b = x^{a+b}$

$2^3 \times 2^2 = 2^{3+2}$

$$M_{\alpha}(t) = M_{\alpha}(ct)$$

$$M_{\alpha}(t) = E[e^{cxst}]$$

$$= E[e^{c(t)x}]$$

$$M_{\alpha}(t) = M_{\alpha}(ct)$$

at $M_{\alpha}(t)$